

Computer Networks - Short Answers

Q1. 1. Segments= $150 \times 1000 \times 1000 / 1500 = 100000$. Frames=100000. Overhead=4,000,000 bytes (~4 MB). Time= $(154,000,000 \times 8) / 50 \times 10^6 \approx 24.64$ s.

Q2. 2. Windows= $48 / 6 = 8$. Retransmissions=8.

Q3. 3. Payload/frag=1480. Fragments=9. Overhead= $9 \times 20 = 180$ B.

Q4. 4. Same as Q2.

Q5. 5. Checkpoints before failure=6. Retransmit=30 MB.

Q6. 6. Compressed=540 MB. Encrypted=567 MB. Reduction=37%.

Q7. 7. Requests=1024. Time ≈ 214.7 s.

Q8. 8. Total delay=45 ms.

Q9. 9. Frames=50000. Overhead=4,000,000 B. Efficiency=95%.

Q10. 10. Compressed=168 MB. Segments=120000. Overhead=7,200,000 B. Final ≈ 175.2 MB.